UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Short notes and keywords

Short answers, a keyword index, rapid recall and a last-10-minutes page.

Class 011 (Legendre) · Legendre's and Cauchy's linear differential equations · compiled 17 October 2026

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1 Short notes

1 · Linear equations with variable coefficients: Legendre's and Cauchy's forms
Legendre: $k_0(ax+b)^n y^{(n)} + k_1(ax+b)^{n-1}y^{(n-1)} + \cdots + k_n y = X$; Cauchy: the same with x^k ($a=1$, $b=0$). Variable coefficients, but the degree of each coefficient equals the order of its derivative, so a substitution reduces it to constant coefficients. $xy'''' + 10y''' = 0$ becomes Cauchy after multiplying by x^3.

2 · The substitution $ax+b=e^z$ and the operator results
$ax+b=e^z$, $z=\log(ax+b)$, $\frac{dz}{dx}=\frac{a}{ax+b}$, $D=\frac{d}{dz}$: $(ax+b)y' = aDy$, $(ax+b)^2y'' = a^2D(D-1)y$, $(ax+b)^3y''' = a^3D(D-1)(D-2)y$. Write X in z ($x=\frac{e^z-b}{a}$), find CF and PI in z, convert back ($e^{mz}=(ax+b)^m$). Cauchy: $x=e^z$, $xy'=Dy$, $x^2y''=D(D-1)y$.

3 · Legendre problems: the slide example, CW P29–P30 and five ESA questions
Slide: $(2x+3)^2y''-2(2x+3)y'-12y=6x \to (D^2-2D-3)y=\frac{3}{4}(e^z-3)$, $y=c_1(2x+3)^3+\frac{c_2}{2x+3}+\frac{3}{4}-\frac{3(2x+3)}{16}$. ESA: Jul-23 $(D-2)^2y=2e^{2z}$, $y=(2x+1)^2[c_1+c_2\log(2x+1)+\log^2(2x+1)]$; Jan s2 $(D-1)^2y=3e^z-2$; Jun-24 $(4D^2-2D-2)y=2e^{2z}+3e^z+4$; Dec-23 $(D^2+1)y=\sin 2z$, $y_p=-\frac{1}{3}\sin(2\log(1+x))$.

4 · Cauchy problems: lecture examples, CW P31–P33 and ESA Jan-24 s3 3d
$x=e^z$: $x^m \to e^{mz}$, $\log x \to z$. CW P31 $y_p=-\frac{x^2}{3}(\log x+\frac{2}{3})$; P32 ($\times x^2$) $y=C_1\log x+C_2+2\log^3 x$; P33 $y_p=\frac{x}{6}+\frac{\log x}{x}$. ESA $x^2y''-4xy'+6y=x^4\sin x$: $(D-2)(D-3)y=e^{4z}\sin e^z$, $y=c_1x^2+c_2x^3-x^2\sin x$.

2 Keyword index

20 terms; the number in brackets is the topic.

$\frac{1}{D-m}X$ (4). $e^{mz}\int Xe^{-mz}dz$, for $e^{4z}\sin e^z$.

$\frac{dz}{dx}$ (2). $\frac{a}{ax+b}$ (Cauchy $\frac{1}{x}$).

$a^2D(D-1)$ (2). $(ax+b)^2\frac{d^2y}{dx^2}$.

$a^3D(D-1)(D-2)$ (2). $(ax+b)^3\frac{d^3y}{dx^3}$.

aD (2). $(ax+b)\frac{dy}{dx}$.

$D=\frac{d}{dz}$ (2). Operator after the substitution.

Back-substitution (2). $e^{mz}=(ax+b)^m$, $z=\log(ax+b)$.

Cauchy's linear equation (1). x^k multiplies the k-th derivative; Legendre with $a=1$, $b=0$.

Cauchy–Euler equation (1). Another name for Cauchy's equation.

Chain rule (2). $\frac{dy}{dx}=\frac{dy}{dz}\frac{dz}{dx}$.

Complex roots (3). $\cos(\beta\log(ax+b))$, $\sin(\beta\log(ax+b))$.

Degree = order (1). The recognition test for both forms.

Failure case (3). $f(k)=0$: PI $\frac{ze^{kz}}{f'(k)}$ (in z!).

Legendre's linear equation (1). $(ax+b)^k$ multiplies the k-th derivative.

Multiply through (4). $xy''''+10y'''$ times x^3; CW P32 times x^2.

Repeated root (3). $(ax+b)^m[c_1+c_2\log(ax+b)]$.

RHS in z (3). $x=\frac{e^z-b}{a}$; polynomials give e^{kz} and constants.

Self-learning component (1). Syllabus label of Cauchy's equation; still asked in ESA.

Substitution (2). $ax+b=e^z$, $z=\log(ax+b)$.

Variable coefficients (1). Coefficients that are functions of x.

3 Rapid recall

1	Legendre's equation, second order?	$(ax+b)^2y''+k_1(ax+b)y'+k_2y=X$.
2	Cauchy's equation is the case ...?	$a=1$, $b=0$: $x^2y''+k_1xy'+k_2y=X$.
3	Recognition test?	Degree of each coefficient = order of its derivative.
4	Substitution for Legendre?	$ax+b=e^z$, $z=\log(ax+b)$, $\frac{dz}{dx}=\frac{a}{ax+b}$.
5	$(ax+b)\frac{dy}{dx}$?	aDy, $D=\frac{d}{dz}$.
6	$(ax+b)^2\frac{d^2y}{dx^2}$?	$a^2D(D-1)y$.
7	$(ax+b)^3\frac{d^3y}{dx^3}$?	$a^3D(D-1)(D-2)y$.
8	Cauchy: x^2y''?	$D(D-1)y$ with $x=e^z$.
9	x in terms of z?	$x=\frac{e^z-b}{a}$.
10	e^{3z} back in x for $z=\log(2x+3)$?	$(2x+3)^3$.
11	Slide example's reduced equation?	$(D^2-2D-3)y=\frac{3}{4}(e^z-3)$.
12	Slide example's PI?	$\frac{3}{4}-\frac{3(2x+3)}{16}$.
13	ESA Jul-23 3a in z?	$(D-2)^2y=2e^{2z}$; $y_p=z^2e^{2z}=(2x+1)^2\log^2(2x+1)$.
14	ESA Jan-24 s2 3c in z?	$(D-1)^2y=3e^z-2$; $y_p=\frac{3}{2}(x+2)\log^2(x+2)-2$.
15	ESA Dec-23 3b PI?	$-\frac{1}{3}\sin(2\log(1+x))$ (reading $\sin(2\log(1+x))$).
16	CW P29 PI?	$-\log(1+x)\cos[\log(1+x)]$ (failure).
17	$xy''''+10y'''=0$?	Times x^3; $y=c_1+c_2x+c_3x^2+c_4x^{-7}$.
18	CW P32 answer?	$y=C_1\log x+C_2+2(\log x)^3$ (CW prints $-2(\log x)^2$).
19	ESA $x^2y''-4xy'+6y=x^4\sin x$?	$y=c_1x^2+c_2x^3-x^2\sin x$.

Mnemonic (study aid). *"One factor per derivative, counting down."* $(ax+b)^ny^{(n)}\to a^nD(D-1)\cdots(D-n+1)$: n factors, starting at D and stepping down by 1 (study aid).

4 Last 10 minutes before the exam

Top facts

1. Legendre: $(ax+b)^k$ with the k-th derivative; Cauchy: x^k ($a=1$, $b=0$). Asked in all 8 ESA papers (6–7 marks).
2. $z=\log(ax+b)$: $(ax+b)y' = aDy$, $(ax+b)^2y'' = a^2D(D-1)y$, $(ax+b)^3y''' = a^3D(D-1)(D-2)y$, $D=\frac{d}{dz}$.
3. Rewrite the RHS with $x=\frac{e^z-b}{a}$; solve in z; return with $e^{mz}=(ax+b)^m$, $z=\log(ax+b)$.
4. Watch failure cases in z: $\frac{1}{(D-k)^2}e^{kz}=\frac{z^2}{2}e^{kz}$ (Jul-23, Jan-24 s2), $\frac{ze^{kz}}{f'(k)}$ (Jun-24, CW P29, P30, P33).
5. Constant on the right: $\frac{c}{f(0)}$.
6. ESA Jul-23: $y=(2x+1)^2[c_1+c_2\log(2x+1)+\log^2(2x+1)]$. ESA Jan-24 s3: $y=c_1x^2+c_2x^3-x^2\sin x$.
7. CW P31, P32 printed answers are wrong: $-\frac{x^2}{3}(\log x+\frac{2}{3})$, $2(\log x)^3$.

Top formulas

E2.1 · The substitution and $\frac{dz}{dx}$ ESA ★★★

$$ax+b=e^z, \qquad z=\log(ax+b), \qquad x=\frac{e^z-b}{a}, \qquad \frac{dz}{dx}=\frac{a}{ax+b}$$

GIVES	The change of variable.
APPLIES	Every Legendre equation.
NOTE	Cauchy: $x=e^z$, $z=\log x$, $\frac{dz}{dx}=\frac{1}{x}$.

S011L·5 · T011L 00:05:20 · T011L 00:06:11 · T011Lb 00:01:49

E2.2 · First derivative ESA ★★★

$$\frac{dy}{dx}=\frac{dy}{dz}\cdot\frac{a}{ax+b} \quad\Rightarrow\quad (ax+b)\frac{dy}{dx}=aDy, \qquad D=\frac{d}{dz}$$

GIVES	The first-order term.
APPLIES	Chain rule.
NOTE	Cauchy: $x\frac{dy}{dx}=Dy$.

S011L·6 · T011L 00:07:16 · T011L 00:07:44 · T011L 00:18:54

E2.3 · Second derivative ESA ★★★

$$(ax+b)^2\frac{d^2y}{dx^2} = a^2D(D-1)y = a^2(D^2-D)y$$

GIVES The second-order term.

APPLIES Product rule on $\frac{a}{ax+b}\frac{dy}{dz}$, then the chain rule.

NOTE Cauchy: $x^2\frac{d^2y}{dx^2} = D(D-1)y$. Not a^2D^2y.

S011L·6 · T011L 00:08:53 · T011L 00:11:45 · T011L 00:19:07

E2.5 · Second-order Legendre equation reduced ESA ★★★

$$(ax+b)^2y'' + k_1(ax+b)y' + k_2y = X$$
$$\implies \left(a^2D(D-1) + k_1aD + k_2\right)y = X\left(\tfrac{e^z-b}{a}\right)$$

GIVES A constant-coefficient equation in z.

APPLIES Every ESA question.

NOTE Divide by a^2 if convenient; the AE is $a^2m(m-1) + k_1am + k_2 = 0$.

S011L·7 · T011L 00:14:20 · T011L 00:15:57 · ESA Jul-23 3a

E2.6 · Back to x ESA ★★★

$$e^{mz} = (ax+b)^m, \qquad z = \log(ax+b), \qquad \cos\beta z = \cos(\beta\log(ax+b))$$

GIVES The final answer in x.

APPLIES Last step, always.

NOTE Repeated root m: $(c_1+c_2z)e^{mz} = (ax+b)^m[c_1 + c_2\log(ax+b)]$.

S011L·9 · T011Lb 00:09:11 · T011Lb 00:12:12 · T011Lb 00:17:39

E3.2 · PI rules needed in z (classes 003–004) ESA ★★★

$$\frac{1}{f(D)}e^{kz} = \frac{e^{kz}}{f(k)},\ f(k)\neq 0; \qquad \frac{1}{f(D)}e^{kz} = \frac{z\,e^{kz}}{f'(k)},\ f(k)=0\neq f'(k)$$
$$\frac{1}{(D-k)^2}e^{kz} = \frac{z^2}{2}e^{kz}, \qquad \frac{1}{D^2+\beta^2}\sin\beta z = -\frac{z}{2\beta}\cos\beta z$$

GIVES Every PI in Topic 3.

APPLIES Failure cases in 4 of 6 ESA questions.

NOTE A constant c is ce^{0z}: $\frac{c}{f(0)}$.

ADDED FOR CLARITY T011Lb 00:10:29 · T011Lb 00:16:27 · ESA Jul-23 3a · ESA Jan-24 s2 3c · CW U2 P29